



FSPM0034 Test 2: Statistics and Probability Examination - Semester 1, 2022/2023

Statistics and Probability (Universiti Teknologi Malaysia)



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TEST 2

SEMESTER I, SESSION 2022/2023

COURSE CODE : FSPM0034

COURSE : STATISTICS AND PROBABILITY

PROGRAMME : FOUNDATION

DURATION : 90 MINUTES

DATE : 30th JUNE 2022

TIME : 8:15 PM – 9:45 PM

INSTRUCTION TO CANDIDATES:

1. This paper consists of five (5) questions.
 2. All steps must be shown clearly.
 3. Candidates are not permitted to share the answer booklet(s) with anyone.
 4. If one candidate's solution is similar with other candidates, the marks will be shared among them. For example, if the question worth five marks and there are two candidates who answered it correctly and identically, each candidate will get half out of five marks.
 5. Candidates can be selfish in this situation by not sharing answers with other candidates.
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WARNING!

Students caught copying/cheating during the examination will be liable for disciplinary actions and SPACE may recommend the student to be expelled from the study.

This examination question consists of (5) printed pages only including this page

**Foundation Programme
Universiti Teknologi Malaysia
Session 2022/2023 Semester 1**

Test 2 (20%)

Time: 90 minutes

FSPM0034 Statistics and Probability

Answer all questions. Show all your work clearly.

1. The probability function of a random variable X representing the number of people entering Fitness Gym from 5:00 to 6:00 p.m. on a particular day is given as follows:

$$f(x) = p \left(\frac{x+1}{2} \right), \quad x = 2, 3, 4, 5;$$

where p is a constant.

- i. Find the value of p and construct the cumulative distribution table of X , $F(x)$.
(3 marks)
- ii. Sketch the graph of $F(x)$.
(2 marks)
- iii. Find $P(|X| > 3)$.
(3 marks)

2. The reaction time (in seconds) to a certain stimulus is a continuous random variable with probability density function,

$$f(w) = \begin{cases} \frac{k}{2w^2}, & 1 \leq w \leq k \\ 0, & \text{otherwise} \end{cases}$$

where k is a constant.

- i. Show that the value of k is 3.
(2 marks)

ii. Compute $P(0 < W < 1.5)$.
(3 marks)

iii. Find the values of a and b if the probability that reaction time is between a and b is $\frac{2}{5}$ and the probability that reaction time exceeds a is $\frac{1}{2}$.
(5 marks)

3. A continuous random variable X has probability density function as follows:

$$f(x) = \begin{cases} \frac{3}{5}, & 0 \leq x < 1 \\ \frac{3}{20}(x-3)^2, & 1 \leq x \leq 3 \\ 0, & \text{otherwise} \end{cases}$$

Based on the given probability density function, find:

i. the expected value of X ,
(2 marks)

ii. the value of c if given $E(X + c) = 3E(X - c)$,
(2 marks)

iii. $Var(X + c)$.
(3 marks)

4a) An electronic product contains 20 integrated circuits. The probability that any integrated circuit is defective is, m and the integrated circuits are independent. The product operates only if there is no defective integrated circuit.

i. If the average number of defective integrated circuit is 1, what is the probability that any integrated circuit is defective?
(2 marks)

ii. Hence, what is the probability that the product operates?
(2 marks)

- 4b) 85% of a large lot of electrical components are non-defective. Fifteen batches of twenty components each are drawn from this lot at random.
- i. A batch is selected randomly, what is the probability that it contains at least fifteen non-defectives electrical components?
(2 marks)
 - ii. What is the probability that more than thirteen batches contain at least fifteen non-defectives electrical components?
(2 marks)
5. In four hours, an average of 10 people visit the Agro Bank.
- a) Find the probability that exactly 8 customers visit the bank during four hours.
(2 marks)
 - b) Given that the bank open 8 hours per day. Find the probability on a given working day, the number of customers visit the bank is:
 - i. more than 12,
(3 marks)
 - ii. between 11 to 15.
(2 marks)

END OF QUESTION PAPER

LIST OF FORMULA**Random Variables**

$$E(X) = \sum xP(X = x)$$

$$E(X) = \int xf(x)dx$$

$$Var(X) = E(X^2) - [E(X)]^2$$

Probability Distributions

Distribution	Probability Functions	Mean	Variance
Binomial	$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x},$ where $x = 0, 1, \dots, n$ and $0 \leq p \leq 1$	np	$np(1-p)$
Poisson	$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!},$ where $x = 0, 1, 2, \dots$ and $\lambda > 0$	λ	λ
Normal	$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right),$ where $-\infty < x < \infty, -\infty < \mu < \infty$ and $\sigma > 0$	μ	σ^2
Standard Normal	$Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$	μ	σ^2